

CD-A Niche 4 Domain-KL 100-Mode Inversion

Three-Hour Native OGS Campaign

Generated from `inversion_attempt`

2026-06-08

Executive Summary

This report records the completed 365-day native OGS campaign `ert10_support_365d_domain_kl_100_3h_native_s1_v1`. The campaign ran 16 parallel Metropolis-Hastings chains with a 100-mode 2D domain-KL intrinsic-permeability magnitude field. The run completed cleanly, wrote `campaign_samples.csv` and `campaign_manifest.json`, and left no live OGS worker processes for this campaign.

The best ERT candidate is `chain06_step0039_proposal`. Its fitted open-niche hydraulic-boundary time shift is **+89.035 days**. The 10-point ERT objective is 0.278451 and the 10-point mean absolute residual is 0.071674 in log10 resistivity. On the independent 100-point circular ERT diagnostic, the mean annual MAE is 0.101334 log10, compared with 0.111878 for the previous rough-RF baseline, improving 62 of 100 circular supports.

1 Campaign Status

Quantity	Value
Campaign rows	1221
Manifest wall time	10912.413 s
Requested max wall time	10800.0 s
Horizon	365 days
Backend	native
Parallel chains	16
Best ERT candidate	<code>chain06_step0039_proposal</code>

Table 1: Completed campaign status. The small wall-time overrun is expected because the runner waits for in-flight OGS calls before writing final files.

2 Setup And Smoke Gates

Before the 3-hour campaign, the new 100-mode domain-KL field was fitted to the previous rough-RF best field and smoke-tested through native OGS. The KL coefficient gate was $|\xi| < 6$; the fitted approximation passed with maximum $|\xi| = 3.78037$ and p95 $|\xi| = 1.53246$. The native smoke run had ERT objective 0.364329 and ERT MAE 0.090837.

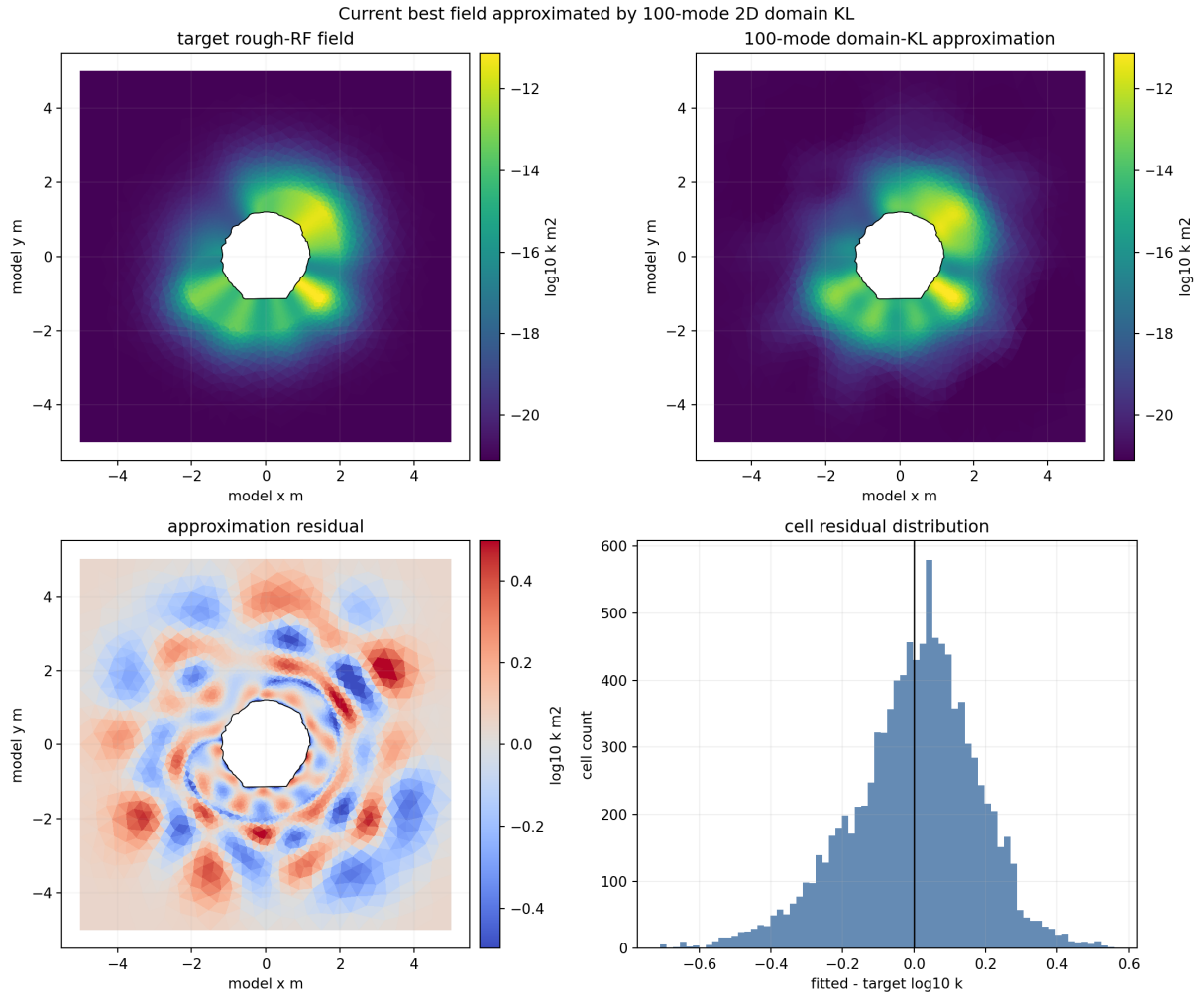


Figure 1: Domain-KL approximation of the previous rough-RF best permeability field before the full campaign.

3 Parameterization And Priors

The active OGS field is tensor-valued intrinsic permeability `k_i_rd`. The domain-KL random field controls only the scalar `log10` magnitude. The tensor written to OGS is

$$K_i(x) = R(\alpha) \left(10^{k_0 + \sigma_f \sum_{j=1}^{100} \xi_j \sqrt{\lambda_j} \phi_j(x)} \begin{bmatrix} 1 & 0 \\ 0 & q \end{bmatrix} \right) R(\alpha)^T.$$

Here k_0 is the global `log10` intrinsic-permeability level, $\sigma_f = 5.0$ is the domain-field standard deviation, ϕ_j are the 2D KL basis functions computed on mesh cell centroids, α is bedding angle, and $q = K_{22}/K_{11}$ before rotation.

Parameter	Prior or fixed value	Meaning
k_0	Normal(-16, 3)	global <code>log10</code> magnitude level
$\xi_1, \dots, \xi_{100}$	iid Normal(0, 1)	KL coefficients
σ_f	5.0 fixed	field amplitude scale
Correlation length	0.45 m fixed	Wendland C2 covariance scale
Compact support radius	1.35 m fixed	sparse covariance support
α	Normal(144 deg, 5 deg)	bedding rotation angle
q	Uniform(0.1, 0.9)	local K_{22}/K_{11} ratio

Table 2: Permeability-field priors and fixed KL settings.

4 Best ERT Fit

Quantity	Value
ERT objective	0.278451
ERT 10-support MAE	0.071674 log10
Log posterior	-2.960956
Log likelihood	26.265375
Log prior	-29.226331
Boundary shift	+89.035430 days
Archie $\log_{10} a$	0.785779
Archie exponent b	-0.738772
k_0	-21.003932 log10 m2
Bedding angle	140.687576 deg
q	0.431179
Max $ \xi $	4.059912
p95 $ \xi $	1.290767

Table 3: Best ERT candidate parameters and likelihood scores.

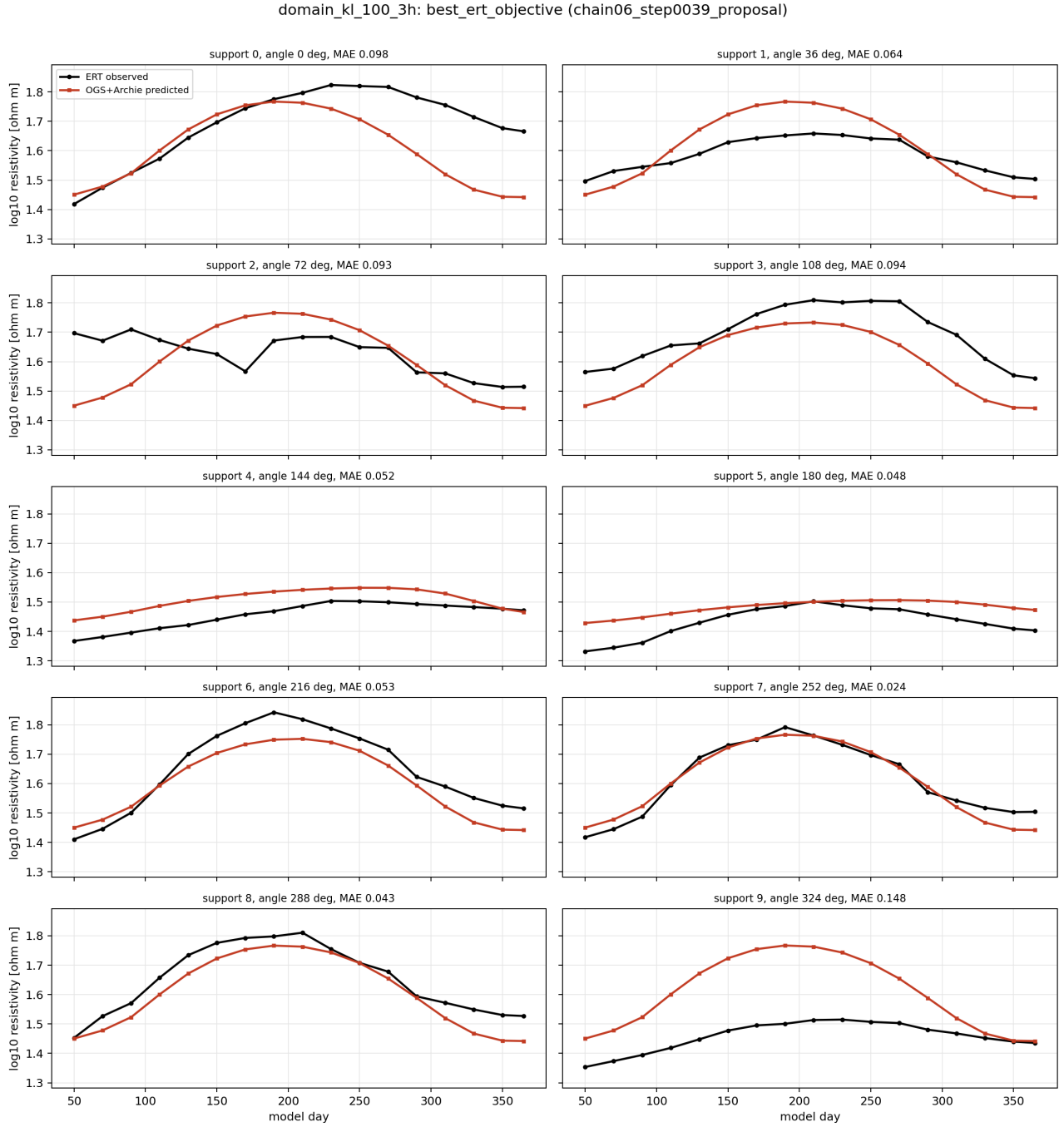


Figure 2: First-year ERT evolution at the 10 support disks: observations against OGS+Archie predictions.

5 Scalar Parameter Posteriors

The posterior summary below uses the empirical MH state distribution in `campaign_samples.csv`, excluding the 16 `resume_state` rows. It excludes the 100 KL coefficients and cellwise permeability-field summaries, but keeps the scalar parameters that control the tensor level/orientation, Archie transform, porosity, NMR offset, ERT texture floor, and boundary curve. Because this was a short 3-hour campaign, these are exploratory posterior summaries rather than a convergence claim. The boundary suction scale and offset were fixed in this campaign, so their posterior rows are degenerate. The histogram panels are normalized as empirical posterior density estimates, with KDE curves and prior PDFs overlaid for direct comparison. The two boundary suction panels are zoomed fixed-value panels rather than sampled distributions.

Parameter	mean	std	p05	median	p95	best ERT
k0 log10 k [m2]	-20.94	0.1908	-21.23	-20.97	-20.67	-21
bedding angle [deg]	141.6	1.027	140.1	141.5	143.7	140.7
q = K22/K11	0.4234	0.03468	0.3612	0.4286	0.4751	0.4312
homogeneous porosity	0.1288	0.008242	0.1186	0.1259	0.1459	0.1446
Archie log10 a	0.7307	0.0951	0.57	0.7549	0.8778	0.7858
Archie exponent b	-0.7428	0.08711	-0.8868	-0.7337	-0.6078	-0.7388
NMR bound water	0.04788	0.006391	0.03494	0.04873	0.05783	0.05352
ERT texture floor log10	0.1603	0.05592	0.08355	0.153	0.2719	0.2837
boundary shift [d]	105.1	17.11	77.96	105.4	132.6	89.04
boundary suction scale	1	0	1	1	1	1
boundary suction offset [MPa]	0	0	0	0	0	0

Table 4: Scalar parameter posterior summaries from 1205 non-resume MH states. The best-ERT column marks `chain06_step0039_proposal`.

Scalar MH posterior density estimates with prior PDFs, excluding 100 KL field coefficients

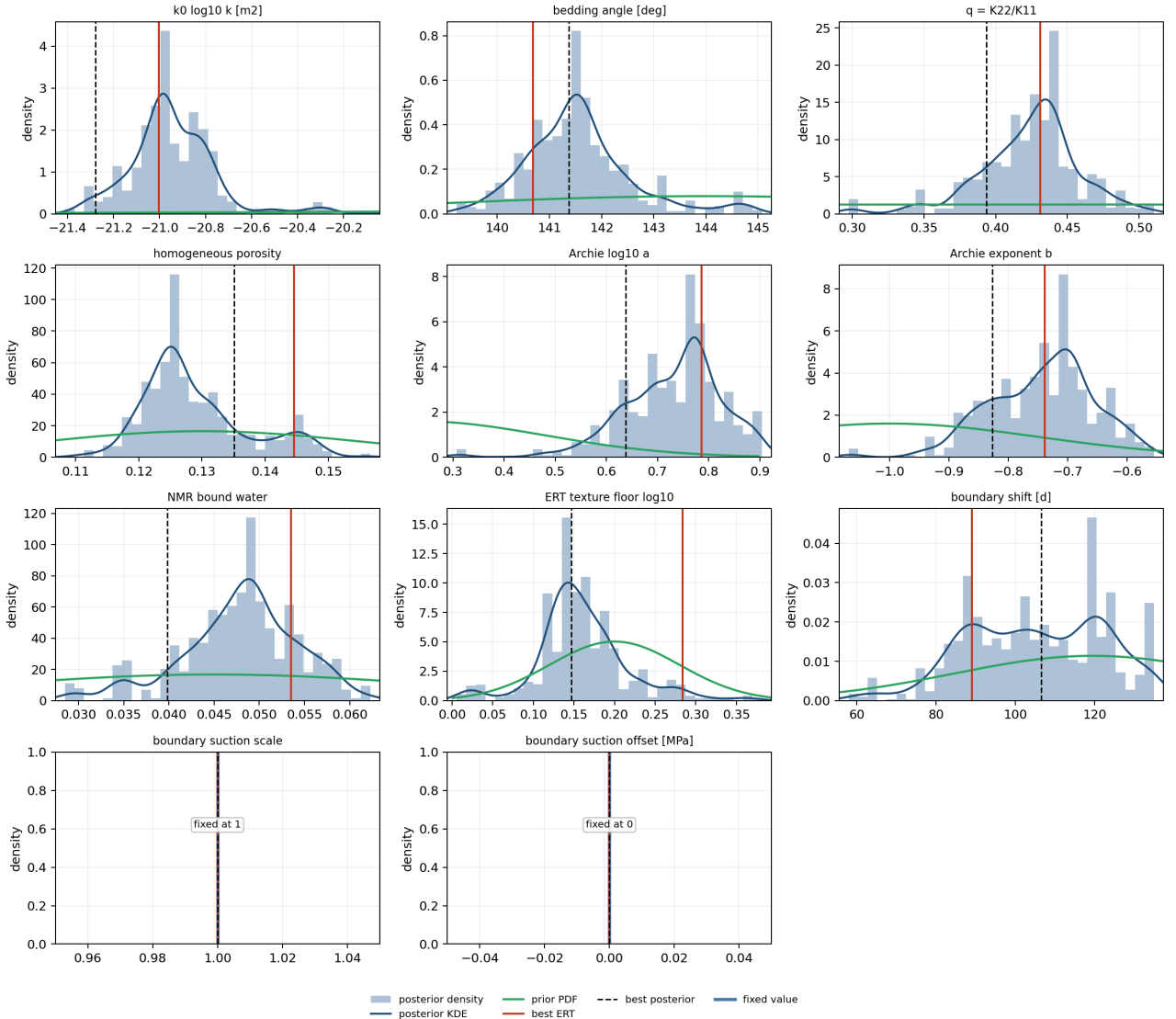


Figure 3: Empirical posterior density estimates for scalar parameters. Blue bars are normalized posterior densities, dark-blue curves are KDE estimates, green curves are prior PDFs, red lines mark the best ERT candidate, and dashed black lines mark the best posterior candidate. Fixed boundary suction scale and offset are shown as zoomed fixed-value panels.

Scalar MH traces by chain, excluding 100 KL field coefficients

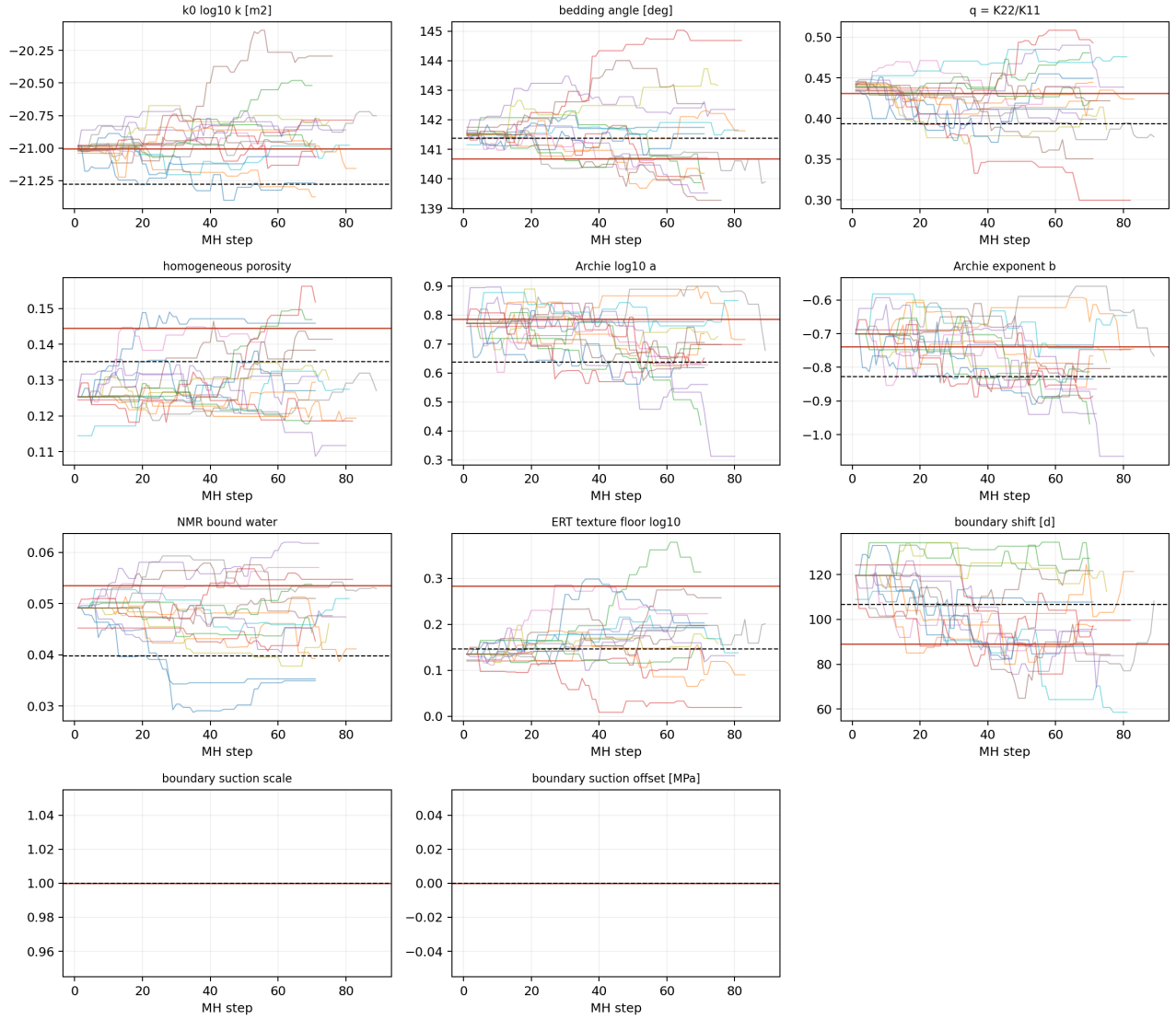


Figure 4: Scalar parameter MH traces by chain. Red horizontal lines mark the best ERT value; dashed black lines mark the best posterior value.

6 Permeability Field

chain06_step0039_proposal | q=0.431, angle=140.69 deg, ERT objective=0.2785

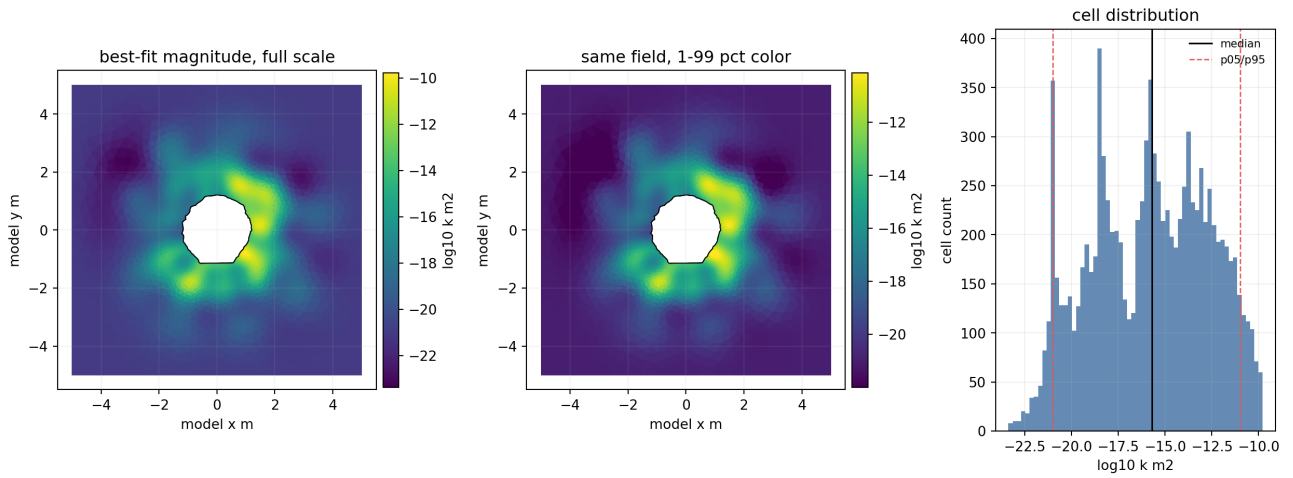


Figure 5: Best-fit scalar intrinsic-permeability magnitude recovered from tensor $\mathbf{k}_{i-\mathbf{rd}}$ as $\text{trace}(\mathbf{K}_i)/(1+q)$.

Field statistic	log10 m2
Minimum	-23.376174
p05	-21.011958
Median	-15.712623
Mean	-15.851041
p95	-10.987349
Maximum	-9.779248

Table 5: Best-fit scalar permeability magnitude distribution on mesh cells.

7 Circular ERT Error Diagnostics

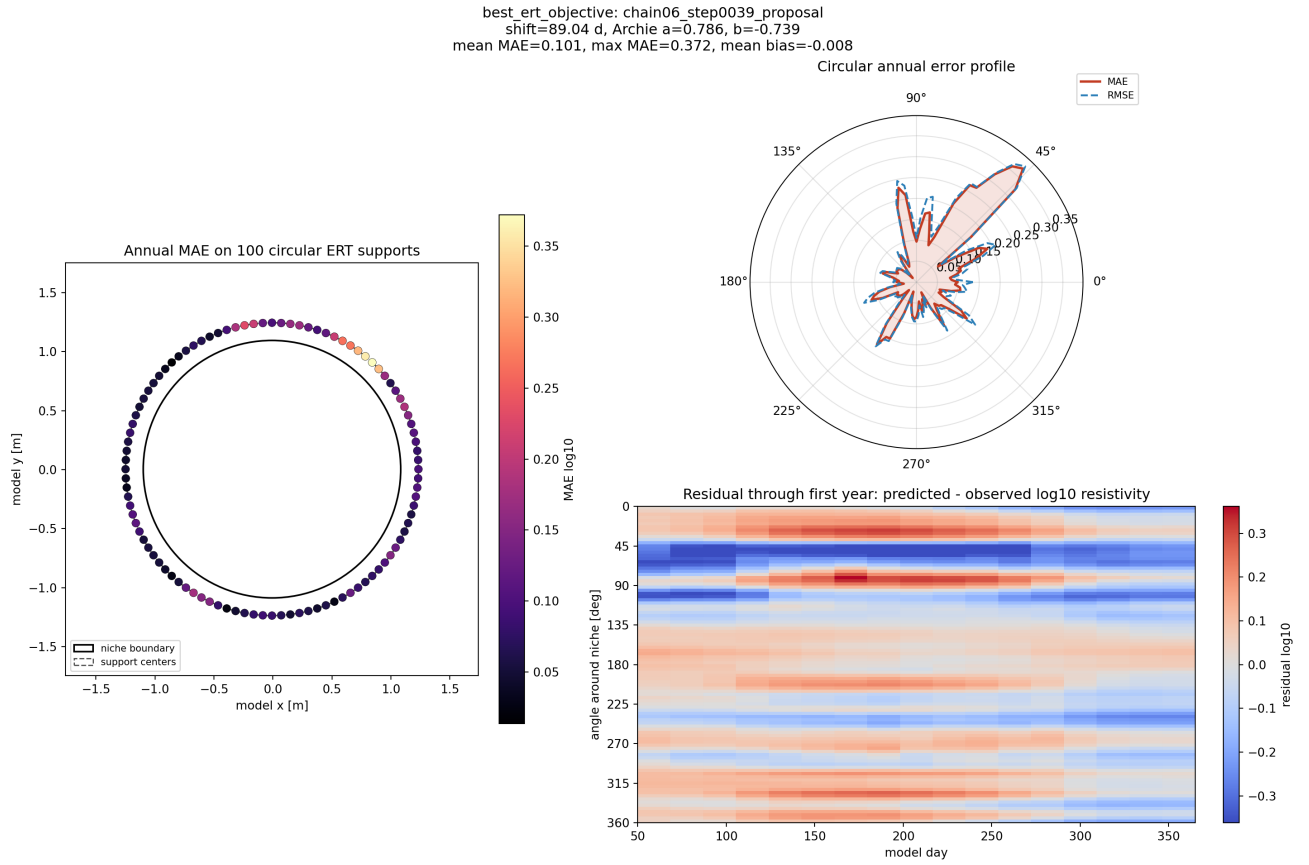


Figure 6: Annual error profile on 100 circular ERT support disks around the niche at approximately 15 cm depth.

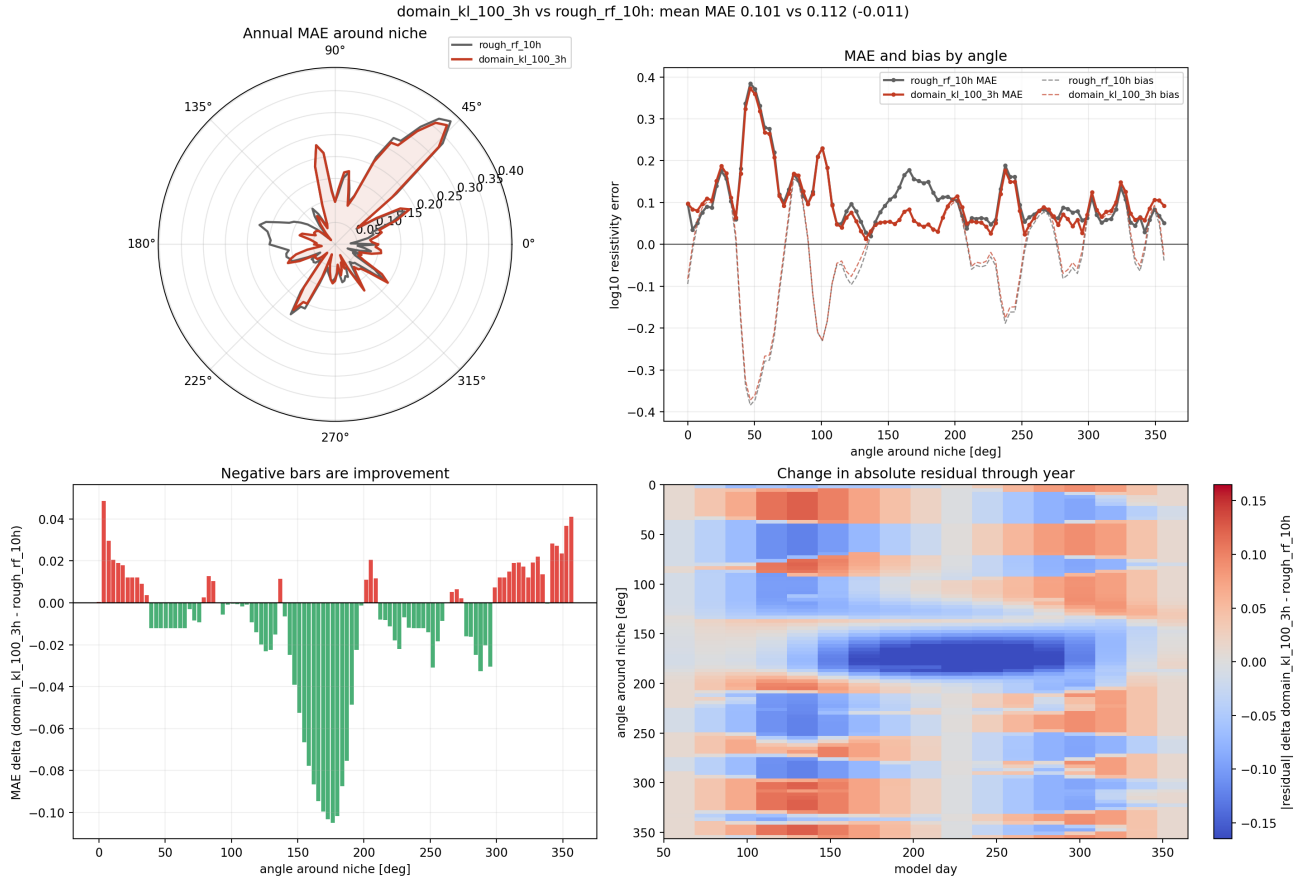


Figure 7: Comparison against the previous rough-RF 10-hour baseline. Negative bars indicate lower MAE for the new domain-KL campaign.

Metric	Rough-RF 10h	Domain-KL 100 3h
Mean radial MAE	0.111878	0.101334
Maximum radial MAE	0.384404	0.372321
Mean MAE change		-0.010545
Improved supports		62 / 100
Worst remaining direction		near 45 deg
Largest improvement direction		176.4 deg

Table 6: 100-point circular ERT diagnostic summary.

8 Commands And Artifacts

The final report bundle was generated with:

```
python3 scripts/write_domain_kl_campaign_report.py \
  --campaign-id ert10_support_365d_domain_kl_100_3h_native_s1_v1 \
  --label domain_kl_100_3h

python3 scripts/plot_domain_kl_scalar_posteriors.py \
  --campaign-id ert10_support_365d_domain_kl_100_3h_native_s1_v1 \
  --label domain_kl_100_3h
```

This PDF was compiled with:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error \
  domain_kl_100_3h_final_report.tex
```

Primary report tables are in **tables/**; final plots are in **figures/**. Raw campaign tables and the Markdown companion remain in the original untracked run outputs.